Product analytics

Understand value. Observe behavior. Test better decisions.

Coverage

A synthesis of the repository's 42-page slide deck, both 10-page worksheets, the one-page IDEO handout and the resource-link files. The PDF includes keyword maps, detailed explanations, worked calculations and interview practice.

How to study

Use the overview to reconstruct the five branches from memory. For each branch, explain its keywords aloud before reading the notes. Answer the interview prompts, then check the answer cues. Revisit the cross-links: the purpose is to explain relationships and make decisions, not only recall definitions.

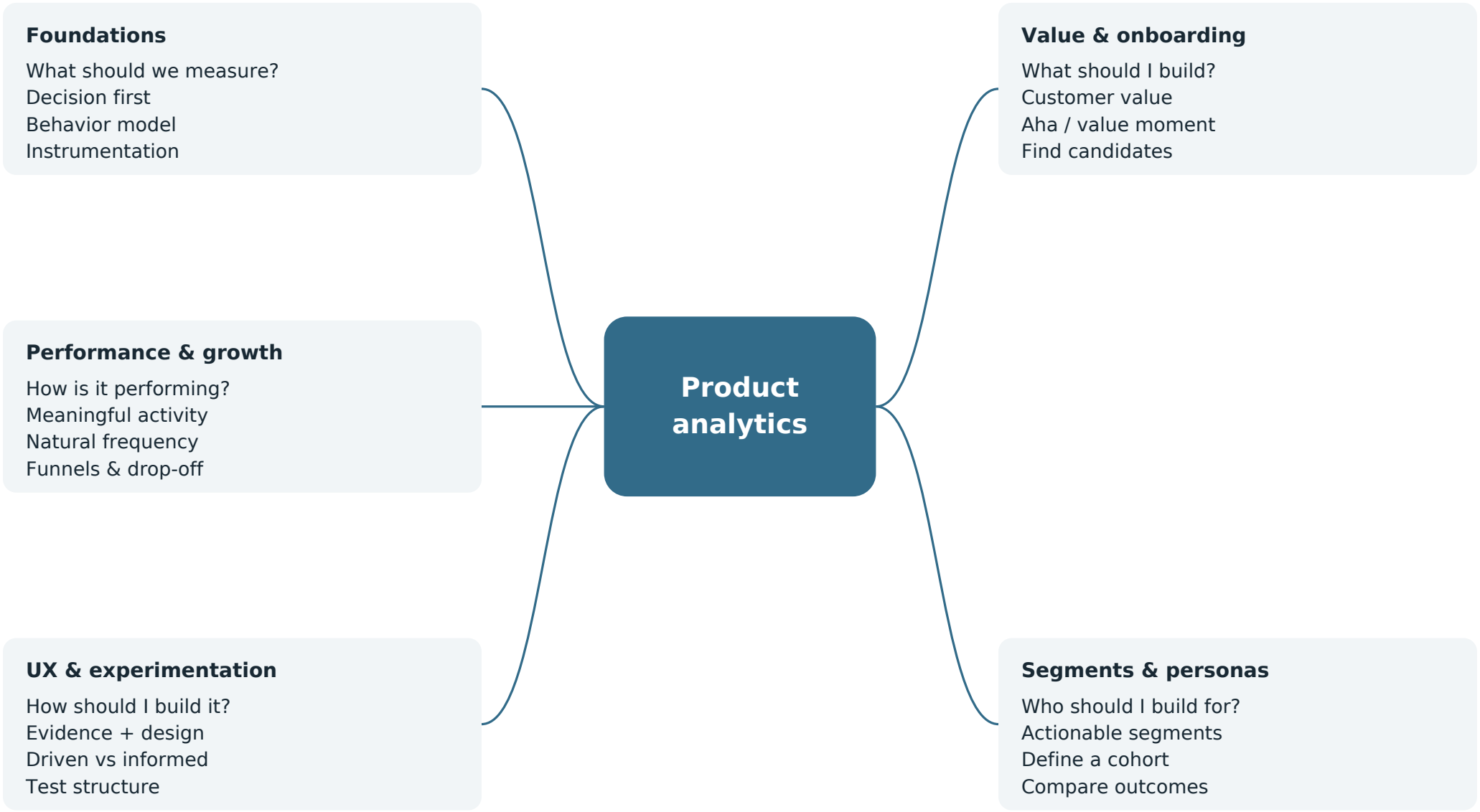
Evidence labels

Source summaries cite PDF page numbers and worksheet exercises. Items marked [extension] add interview precision beyond the course. Historical product examples and Uber Eats thresholds are teaching examples, not verified current company practices or measured experiment results.

Scope

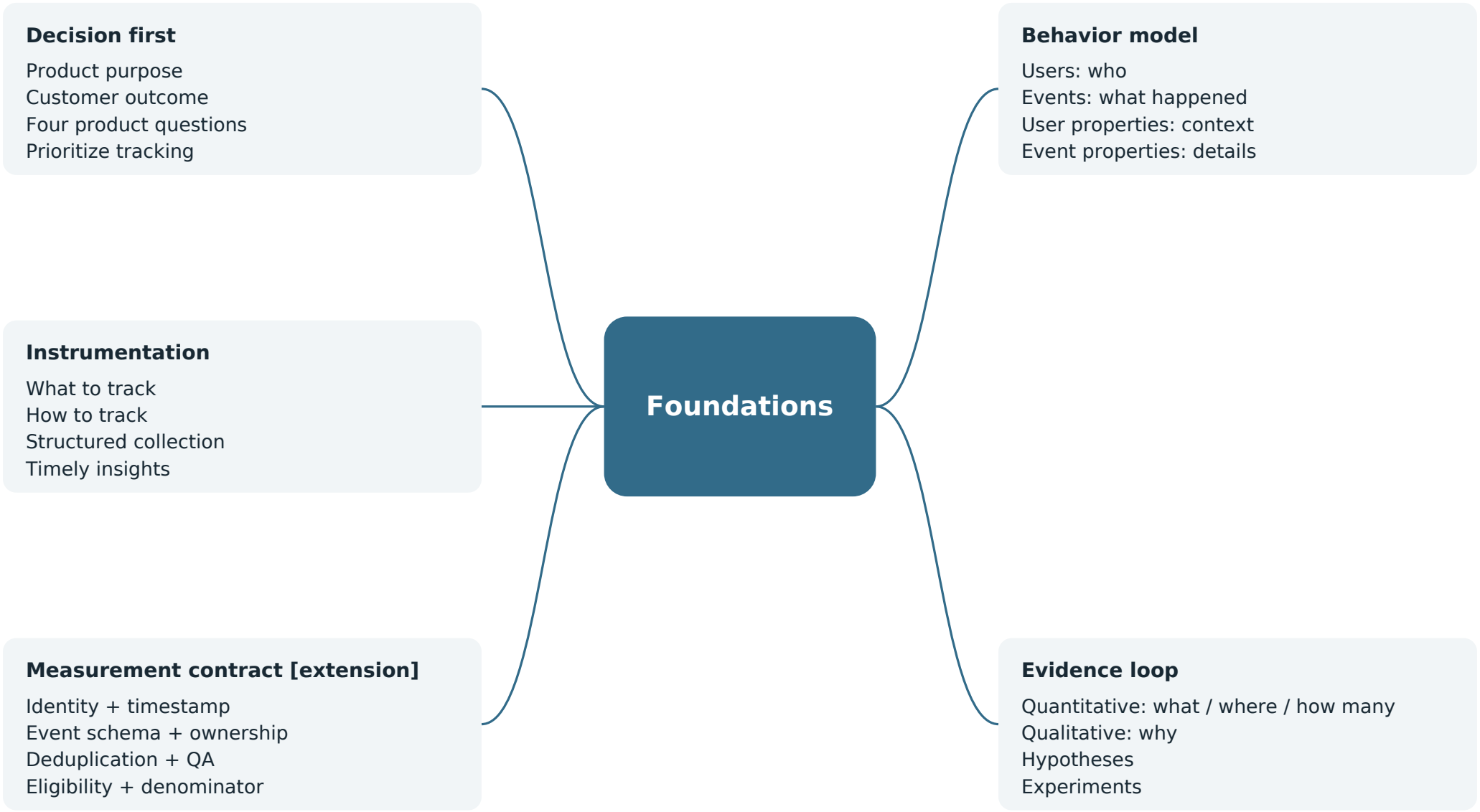
This is a comprehensive guide to the materials present in the repository. It is not a transcript of the video lessons and is not a complete product-analytics interview curriculum. Additional preparation areas are listed at the end.

Product analytics



Overview / five main branches; detailed maps and interview prompts follow

Foundations



Slides pp. 3-5; worksheet exercise 1

What should we measure?

Decision first

Product analytics studies actual product use to inform decisions. Start with the outcome and decision, then decide what evidence is needed. The course connects four questions: what to build, how it performs, whom to build for, and how to build it.

Interview: What would you instrument before launching a new feature?

Answer cue: Clarify the intended customer outcome and business decision; define the behavior and success measure; instrument the minimum events and properties needed to evaluate them.

Behavior model

A user is the actor; an event is an action; user properties describe the actor; event properties describe that occurrence. In the course's party analogy, Jack is a user, shirt color is a user property, drinking is an event, and drink brand or quantity are event properties.

Interview: How do user properties differ from event properties?

Answer cue: User properties characterize the person or account; event properties describe a specific action at a specific time. Preserve historical context when a property changes.

Instrumentation

Instrumentation makes product behavior observable. The course stresses a structured approach to what and how to track, so the resulting information can answer prioritized questions rather than collect activity without a purpose.

Interview: How would you track an ordering journey?

Answer cue: Use meaningful events for restaurant discovery, cart, checkout, order placement and fulfillment, with enough context to locate friction and tie behavior to outcomes.

Evidence loop

Analytics locates patterns and drop-off; interviews and observation investigate explanations. Combine the two to generate a hypothesis and evaluate a change. A behavioral association alone does not establish that the behavior causes retention.

Interview: A dashboard shows a conversion decline. What next?

Answer cue: Validate the measurement, locate the affected stage and segments, investigate user explanations, form hypotheses, and test the most plausible intervention.

Slides pp. 3-5; worksheet exercise 1

What should we measure?

Measurement contract [extension]

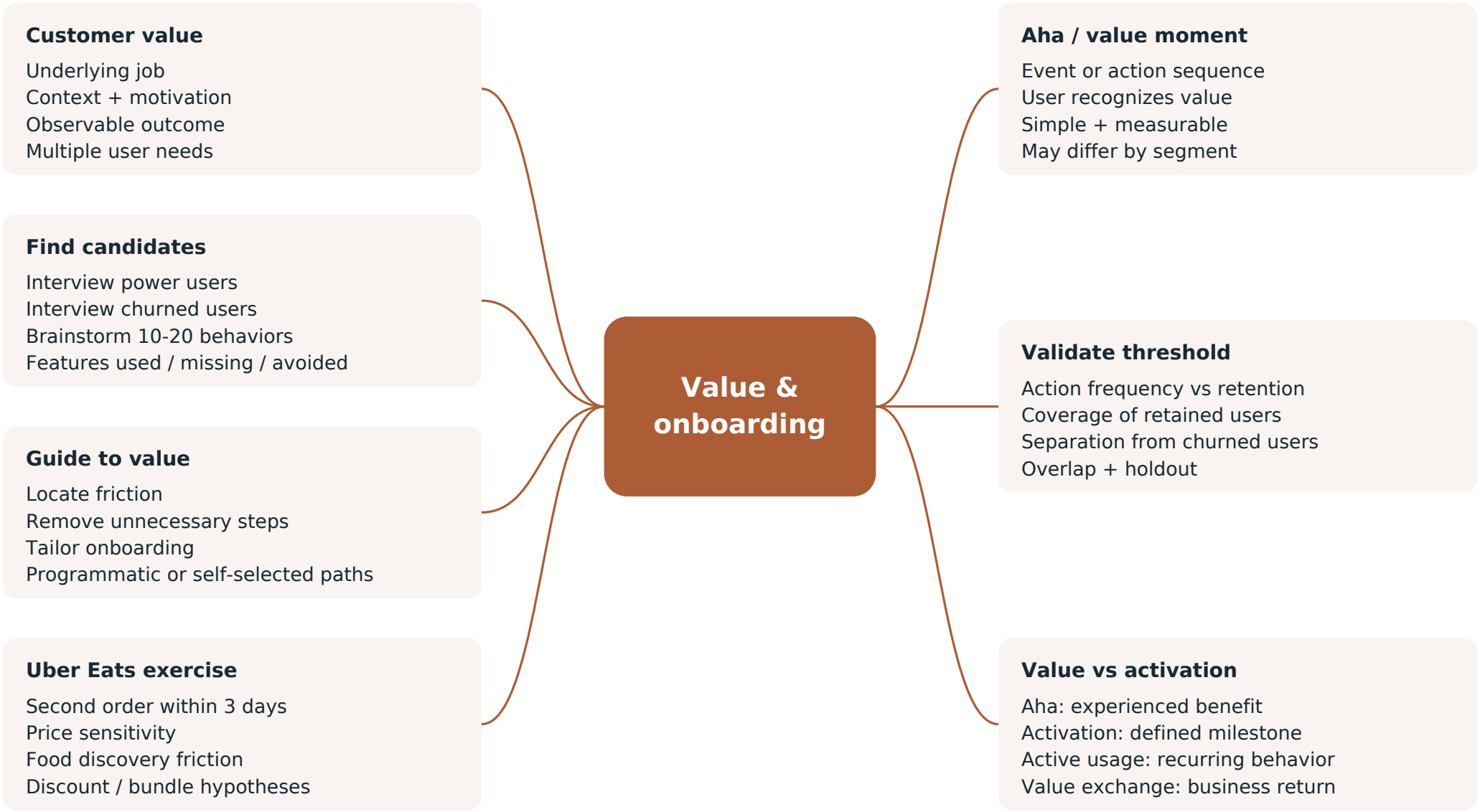
Interview extension: specify stable user/account IDs, event timestamps and time zone, event ownership and schema, duplicate handling, missing events and validation. Every rate needs an explicit eligible population, time window and numerator.

Interview: How could bad instrumentation mimic a product problem?

Answer cue: A missing checkout event can look like abandonment; duplicate orders inflate conversion; broken identity stitching distorts retention. Check tracking before changing the product.

Slides pp. 3-5; worksheet exercise 1

Value & onboarding



Slides pp. 6-17; worksheet exercises 2-4

What should I build?

Customer value

Define the benefit the user seeks before choosing a metric. The same product can deliver different value: motivation, entertainment or support. Translate the underlying need into observable behavior while recognizing that a metric is a proxy for experienced value.

Interview: How would you find a useful activation metric?

Answer cue: Start with the user's job, identify behaviors that plausibly demonstrate value, study retained and churned users, compare later retention, and test interventions.

Aha / value moment

A value moment is an event or sequence associated with the user recognizing the product's benefit. It may occur more than once and may vary by user intent. The course uses historical examples such as adding seven Facebook friends in ten days; treat such thresholds as illustrations, not universal rules.

Interview: Can every user share one value moment?

Answer cue: No. A document tool can deliver value through cross-device access or sharing. A single primary proxy may help focus the team, but distinct jobs can need distinct definitions.

Find candidates

Ask loyal users what they love, use, never use and miss. Ask churned users what outcome they wanted, where value was absent, why they left and what should change. Generate 10-20 candidate behaviors, then inspect analytics rather than selecting the most appealing story.

Interview: Why interview both retained and churned users?

Answer cue: Retained users expose successful paths; churned users reveal unmet needs and friction. Using only enthusiasts introduces survivorship bias.

Validate threshold

The course's four steps are: find important actions, examine retention as usage changes, identify an action-frequency threshold, and test causation. A threshold should both identify retained users and avoid selecting many churned users; an extreme threshold can have impressive retention but cover very few people.

Interview: Why not choose the action with 99% retention?

Answer cue: It may describe a tiny, already-loyal group. Compare population coverage and separation, avoid looking into the future, then test whether encouraging the action improves outcomes.

Slides pp. 6-17; worksheet exercises 2-4

What should I build?

Guide to value

The course highlights friction and unsuitable onboarding as two reasons people leave. Interview users about what they tried and where they got stuck. Personalize onboarding through inferred grouping or user-selected categories; evaluate whether the tailored path helps users reach value and return.

Interview: Would you shorten onboarding for everyone?

Answer cue: Only if evidence supports it. Some steps enable later value, trust or personalization. Test changes by user intent and assess downstream retention, not completion alone.

Value vs activation

The worksheet distinguishes user value from business activation, giving paid subscription as an activation example. Interview clarification: activation commonly means reaching an early value milestone and need not involve payment. Active usage is the recurring qualifying behavior; value exchange is where the business receives value. These can coincide, but often differ.

Interview: Are activation, active usage and monetization identical?

Answer cue: No. Define each for the product. A fitness user can finish a workout, become activated through a first successful routine, return weekly and subscribe at a separate point.

Uber Eats exercise

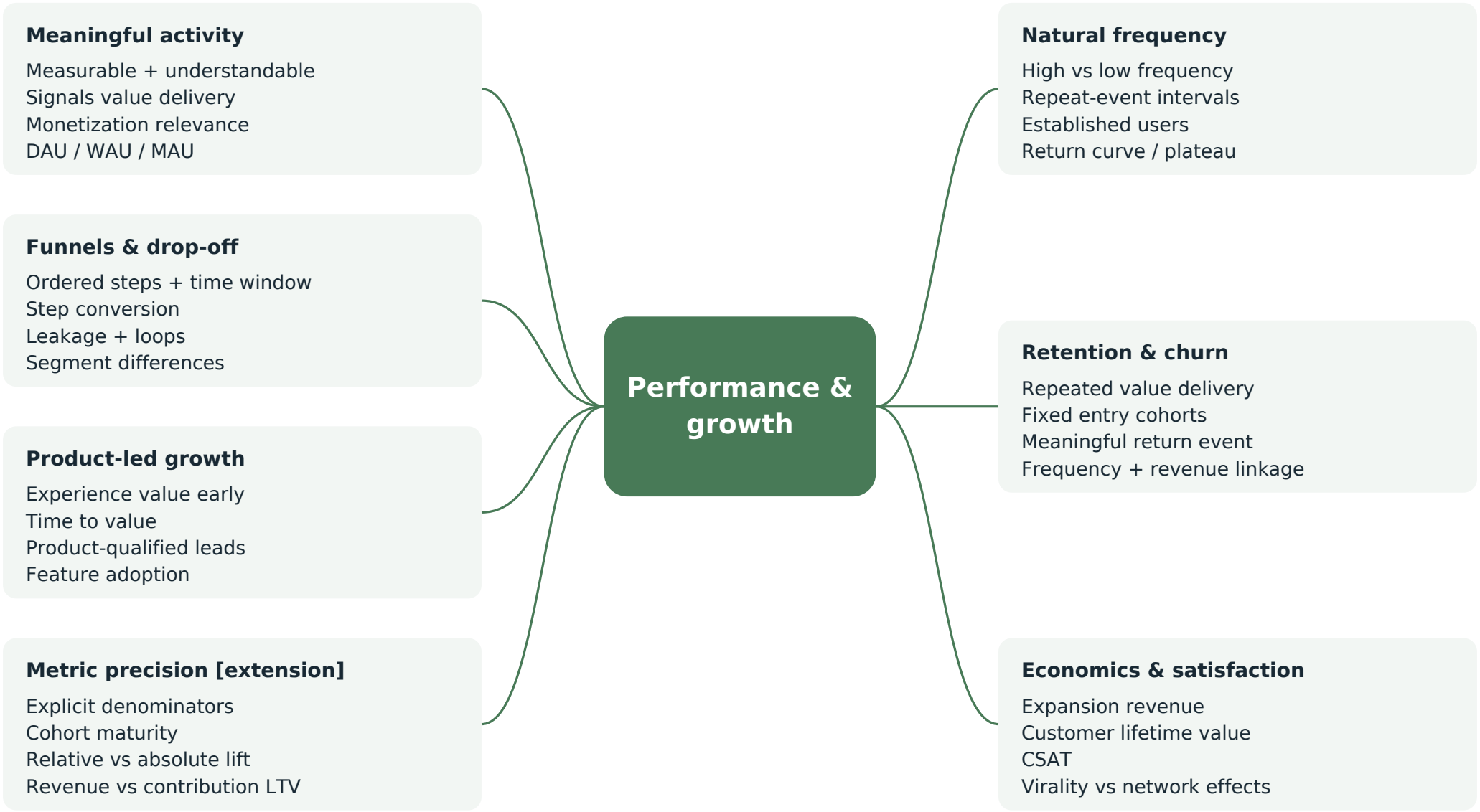
The completed worksheet proposes a second order within three days as the value-moment proxy. It suggests price sensitivity and difficulty finding appealing food as friction. Proposed experiments include restaurant specials, bundles and prepaid meals, with a minimum 2% retention improvement as an exercise criterion. None is presented as a measured result.

Interview: How would you improve this exercise?

Answer cue: Validate the three-day threshold; clarify whether 2% is relative or percentage points; define the retention window; measure incremental retention and contribution margin, not discounted orders alone.

Slides pp. 6-17; worksheet exercises 2-4

Performance & growth



Slides pp. 18-28; worksheet exercise 5

How is it performing?

Meaningful activity

Avoid a definition so loose that opening a page counts as success without meaningful value. The worksheet asks for an active definition that is measurable, understandable and indicative of value delivery and monetization. For Uber Eats, the exercise defines an active user as someone placing at least one order in the period.

Interview: What is an active user for this product?

Answer cue: Name a qualifying value-related event and a justified time window. Count distinct eligible users, then explain how the event relates to customer outcomes and business value.

Natural frequency

Usage expectations must fit the product: messaging may be daily, medical booking much less frequent. In the meditation example, examine established users with at least one session in 60 days, measure time to a second session, compare return windows and inspect the cumulative curve. A plateau suggests a useful descriptive interval, not a proven optimum.

Interview: How would you determine the right retention interval?

Answer cue: Study repeat-value-event timing among sufficiently observed users and segment by use case. Match the interval to the job instead of choosing daily activity by habit.

Funnels & drop-off

A funnel represents the sequence to a value or value-exchange event. Locate the leakiest step, compare converted users' characteristics, look for loops and investigate friction. Pair quantitative drop-off with qualitative research. Define event order, eligibility and time limits before interpreting rates.

Interview: Which funnel stage would you optimize first?

Answer cue: Consider affected users, size of recoverable loss, business impact and feasibility. The largest percentage drop is not automatically the highest-value opportunity.

Retention & churn

Retention asks whether users keep receiving value. Cohorts separate engagement from changing acquisition volume: DAU can rise even while new cohorts retain badly. The course warns against focusing only on revenue, choosing the wrong frequency, or failing to connect retention to revenue. Churn requires an explicit inactivity or contractual definition.

Interview: DAU rose while cohort retention fell. Is the product healthier?

Answer cue: Not necessarily. Acquisition can mask a leaky product. Compare cohort sizes, age-aligned return rates and retained revenue before judging sustainable improvement.

Slides pp. 18-28; worksheet exercise 5

How is it performing?

Product-led growth

Product-led growth uses access to and value from the product as a growth mechanism, contrasted in the course with relying on sales material and demos. Track time to the aha milestone, product-qualified leads from qualifying trial behavior, and feature adoption. Lower friction should support retained value rather than only sign-ups.

Interview: How would you measure a self-serve onboarding improvement?

Answer cue: Track time to value and activation alongside later retention, qualified usage and business outcomes. Guard against shortening time to a shallow milestone that predicts little.

Economics & satisfaction

The course also introduces expansion revenue from existing users, lifetime customer value, customer satisfaction and virality/network effects. Keep the last two distinct: virality describes acquisition through users; a network effect means the product becomes more valuable as relevant participation grows. In a marketplace, participant quality, balance and locality matter.

Interview: Can a product have network effects without virality?

Answer cue: Yes. A product can gain utility from more participants even if acquisition is paid or sales-led. Referral-driven growth alone does not prove a network effect.

Metric precision [extension]

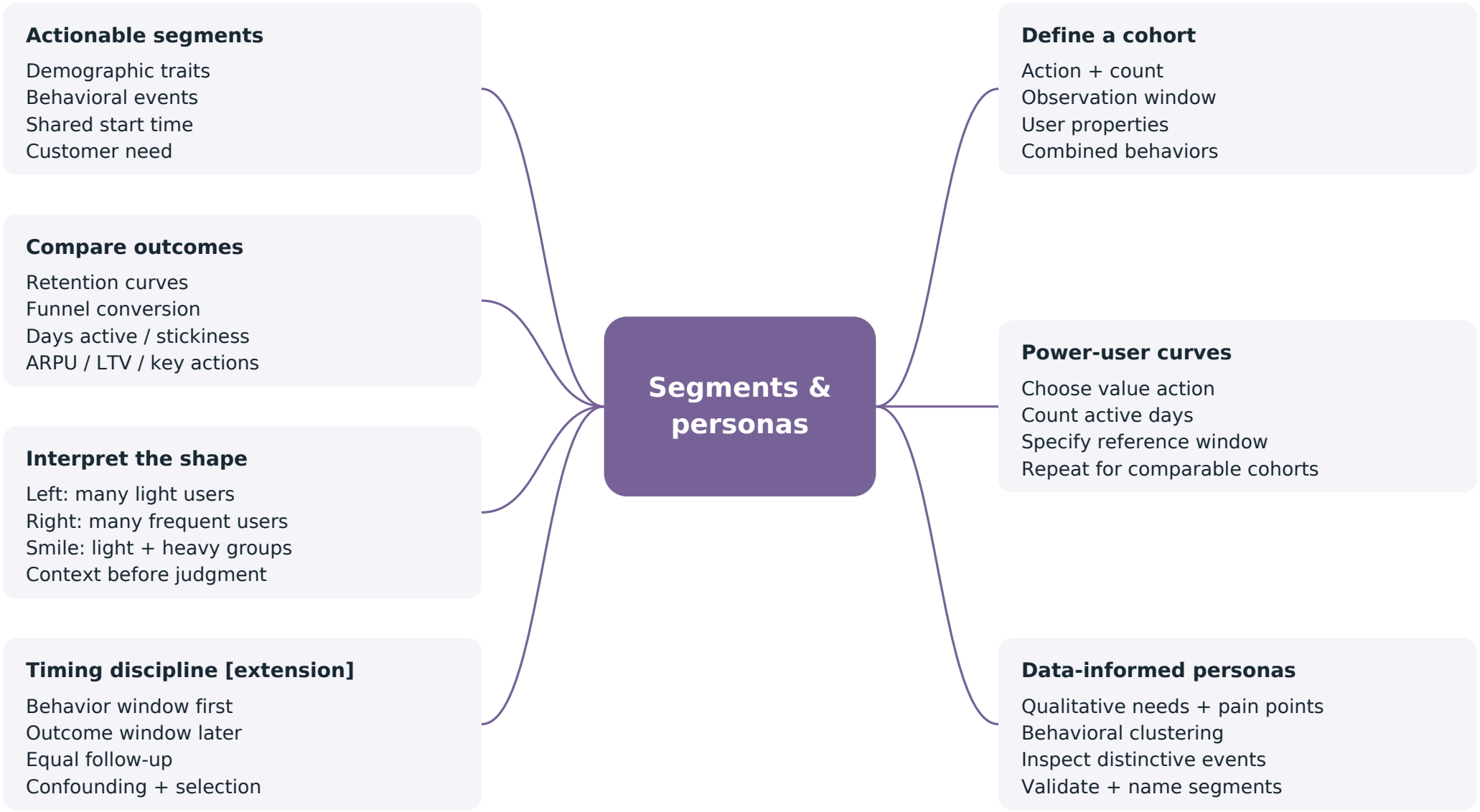
Interview extension: specify exact-day versus rolling retention, feature eligibility, cohort observation length and gross-margin basis for LTV. CSAT usually uses satisfied responses divided by all valid responses; it is not automatically a percentage of all active users. DAU/MAU is a rough usage-frequency proxy, not proof of retention.

Interview: Which metric definitions must be agreed before analysis?

Answer cue: The qualifying event, population, entity, time window, attribution and exclusions. For monetary metrics, also specify currency, net/gross basis and treatment of refunds, discounts and costs.

Slides pp. 18-28; worksheet exercise 5

Segments & personas



Slides pp. 29-35; worksheet exercises 6-8

Who should I build for?

Actionable segments

Segmentation supports a focused product and value proposition. The course contrasts demographic grouping with often more actionable behavioral grouping. Cohorts can share a trait, an event or a time origin. Demographics may still matter when they help explain a real need or constraint.

Interview: When is behavioral segmentation more useful?

Answer cue: When it identifies what people do and a plausible intervention. A need or behavior should explain how the experience could change; a descriptive segment alone does not create a decision.

Define a cohort

Frame the learning question, then define the action, frequency, timeframe, user properties and combined conditions. The worksheet template is: users with specified properties who perform specified actions a stated number of times within a stated window. Make comparison cohorts and timing explicit.

Interview: Define a cohort to study cross-product usage.

Answer cue: For the course example, users with more than five Uber ride bookings in 30 days. Compare later Uber Eats outcomes with a clearly defined comparison group and aligned eligibility.

Compare outcomes

Assess whether a behavior is associated with retention, conversion, usage frequency, revenue and other important actions. The Uber Eats workbook proposes comparing ride-app users and non-users across these outcomes. Such comparisons are observational; differences in intent, location or tenure can explain the association.

Interview: Ride users retain better in Eats. Should you drive everyone to rides?

Answer cue: Not from that result alone. Check comparability, define behavior before the outcome, consider confounding, and test a plausible intervention rather than assuming the relationship is causal.

Power-user curves

A power-user curve shows the distribution of usage frequency, such as the share of users active on one through 30 days in a month. Choose the action and reference window deliberately. Compare distributions over time with consistent definitions; do not confuse this histogram with a retention curve.

Interview: How is a power-user curve different from retention?

Answer cue: It describes frequency within a period; retention describes return after an origin. Both inform engagement but answer different questions.

Slides pp. 29-35; worksheet exercises 6-8

Who should I build for?

Interpret the shape

The course contrasts left-heavy, right-heavy and smile-shaped usage. A left-heavy distribution need not be a failure for an episodic product; the right side need not mean healthy value if activity is burdensome. The Uber Eats exercise hypothesizes a left-heavy curve due to alternatives, low switching costs and first-order incentives.

Interview: Is a smile-shaped curve always good?

Answer cue: It reveals distinct low- and high-frequency groups. Investigate whether heavy use represents value, which segments need help, and whether the natural job supports that frequency.

Data-informed personas

Traditional personas combine research and judgment about goals, experiences, likes and pain points. The course introduces algorithmic behavioral clustering, with an example examining recently active new users and four clusters. Compare characteristic events and interpret each cluster with qualitative research; clusters are not automatically meaningful personas.

Interview: How would you turn clusters into useful personas?

Answer cue: Check stability and interpretability, identify differences in jobs and barriers, talk to users, and tie each persona to a product decision. Do not infer motivations from event counts alone.

Timing discipline [extension]

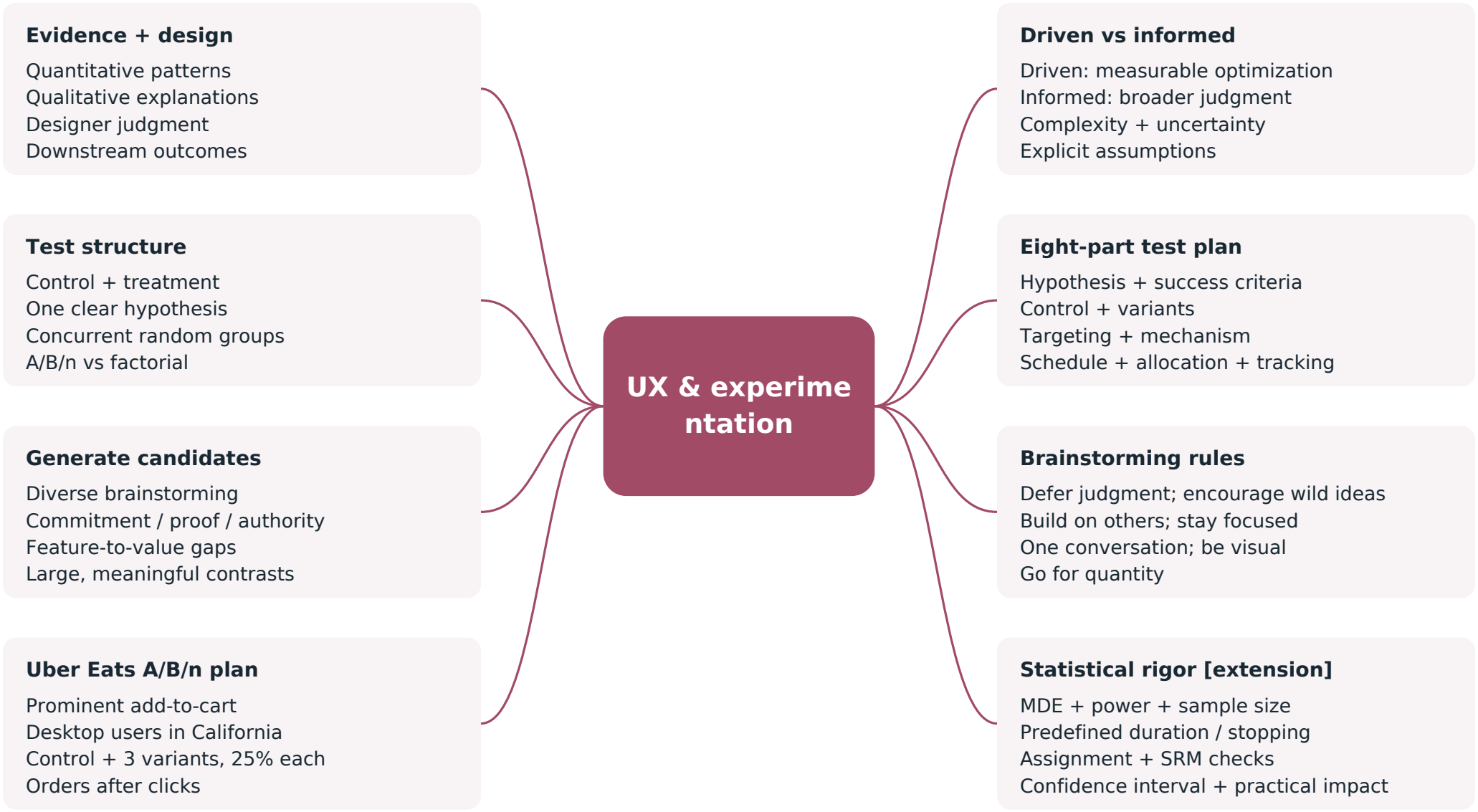
Interview extension: avoid using future behavior to define a cohort and then claiming it predicts earlier retention. Ensure equal opportunity to qualify, adequate follow-up and comparable users. Segment results can differ from aggregates, so inspect mix shifts and avoid fishing through many subgroups.

Interview: What is temporal leakage in cohort analysis?

Answer cue: Information from the outcome period enters the cohort definition. For example, using 30-day activity to predict day-seven retention makes the purported prediction circular.

Slides pp. 29-35; worksheet exercises 6-8

UX & experimentation



Slides pp. 36-42; worksheet exercise 9; IDEO handout

How should I build it?

Evidence + design

UX analytics measures user activity to support design insight. The course emphasizes complementing designers' intuition and knowledge. Use analytics to test an explanation, such as whether a less visible button prevents progress, then examine the downstream outcome rather than click count alone.

Interview: Should analytics make design decisions?

Answer cue: It should inform them. Data can evaluate observable outcomes, while research, accessibility, strategy and design judgment shape which solutions are worth testing.

Driven vs informed

The course favors data-driven choices where the necessary data exists and success can be measured numerically, such as optimization. It favors data-informed judgment for more complex questions with many variables. Neither approach excuses weak evidence or removes the need to define the customer outcome.

Interview: When would you avoid choosing solely by a metric?

Answer cue: When the measure misses material outcomes, the problem is new, or trust, accessibility and long-term strategy matter beyond the observed window. State the trade-off and evidence limits.

Test structure

The slides teach a clear hypothesis, control and treatment, random sampling/assignment and simultaneous variants. They simplify A/B as changing one variable. Clarification: an A/B test can compare coherent packages; it then identifies the package effect. A factorial multivariate test varies factors across combinations to estimate component and interaction effects.

Interview: How do A/B/n and multivariate tests differ?

Answer cue: A/B/n compares several alternative experiences. A factorial test crosses factor levels, for example two headlines by two images, making four combinations; more cells usually require more traffic.

Slides pp. 36-42; worksheet exercise 9; IDEO handout

How should I build it?

Eight-part test plan

The worksheet plan contains eight fields: hypothesis, success criteria, design, targeting, targeting approach, scheduling, traffic allocation and tracking. Mechanisms include URL, cookie, technology, geography, audience profile, query parameter and data-layer targeting. Targeting determines eligibility; random assignment within eligible users determines treatment.

Interview: What belongs in a test plan before implementation?

Answer cue: The causal question, population, variant design, randomization unit, primary and guardrail metrics, measurement, sample-size plan, duration and decision rule. The last statistical safeguards extend the worksheet.

Generate candidates

The course suggests exploring user motivations such as commitment, social proof, liking and authority; mapping features to value drivers; and trying meaningful contrasts, such as video versus text or short versus long experiences. Prefer hypotheses about barriers and value to arbitrary cosmetic changes. Persuasion must preserve truthful information and user choice.

Interview: How would you choose between ten experiment ideas?

Answer cue: Tie ideas to observed friction and a mechanism, assess reachable impact and feasibility, and prioritize tests that can resolve an important uncertainty. Large differences are useful when justified, not for novelty alone.

Brainstorming rules

The IDEO handout's seven principles are to postpone criticism, welcome unusual ideas, develop others' ideas, stay on the question, hold one discussion at a time, sketch ideas, and generate many options. Separate divergent generation from later evaluation against evidence and constraints.

Interview: How do you keep brainstorming useful?

Answer cue: Use a focused challenge and diverse participants, generate options before judging them, make ideas visible, then evaluate their connection to user friction and a testable hypothesis.

Slides pp. 36-42; worksheet exercise 9; IDEO handout

How should I build it?

Uber Eats A/B/n plan

The example compares the existing add-to-cart button with blue, floating and larger versions, allocating 25% to each group. It targets desktop users in California and proposes at least 2% conversion improvement. It lists button clicks and order-confirmation pageviews for tracking. This is an illustrative four-arm plan, with no measured result or sample-size calculation.

Interview: What would you change before running this experiment?

Answer cue: Choose a primary completed-order conversion metric, stable user assignment, order-ID-based tracking, margin and usability guardrails, adequate power and duration, and a multiple-comparison plan. Specify what 2% means.

Statistical rigor [extension]

Interview extension: predefine a minimum detectable effect, significance level, power, sample size and stopping rule. Check sample-ratio mismatch and tracking integrity; use stable randomization; account for multiple variants and peeking. Evaluate uncertainty, effect size, guardrails and business value before rolling out.

Interview: The result is not significant. What can you conclude?

Answer cue: An inconclusive test does not prove no effect. Examine the confidence interval and whether it excludes meaningful benefit or harm; verify power and implementation before deciding whether further work is justified.

Slides pp. 36-42; worksheet exercise 9; IDEO handout

Definitions that survive follow-up questions

Active users

Distinct users who perform the qualifying value-related event within a stated window. DAU: one day; WAU: seven days; MAU: a defined month or rolling window. Declare calendar versus rolling boundaries.

Funnel conversion

Overall conversion = users reaching the final qualifying step / eligible users entering the funnel. Step conversion = users progressing to the next step / users reaching the previous step. Specify order and completion window.

Retention

Period-t retention = cohort users performing the qualifying return event in period t / cohort users eligible for that follow-up. Exact-day, bracketed and rolling/unbounded retention answer different questions; do not interchange them.

Feature adoption

Eligible users using a feature / users eligible to use it. The slides use feature users / total users; restrict the denominator when only some users can access the feature. Define first adoption versus ongoing usage.

Time to value

Elapsed time between a stated origin, such as sign-up, and the value milestone. Report the completion proportion as well as time among completers; a fast median among a tiny successful subset can mislead.

ARPU and LTV

ARPU = revenue in a period / defined user population in that period. Use a matching horizon. LTV estimates lifetime economics; state whether it is revenue or contribution value. Simple average formulas require strong stability assumptions.

Expansion and CSAT

Expansion is additional revenue from existing customers, such as upgrades or more usage. CSAT is typically satisfied valid responses / all valid responses. Describe the survey scale and respondent bias; do not call it satisfaction of all users.

Frequency versus retention

DAU/MAU gives a rough frequency signal under consistent definitions. A power-user curve distributes users by active days. Retention follows return from a shared origin. These measures complement each other; none substitutes for the others.

Four calculations to explain aloud

1 / Conversion and lift

Suppose 1,000 eligible control users generate 200 completed orders, and 1,000 treatment users generate 220. Conversion is 20% versus 22%. Absolute lift is 2 percentage points; relative lift is (22% - 20%) / 20% = 10%. A 2% relative improvement from 20% would instead be 20.4%. These arithmetic differences say nothing yet about statistical reliability.

2 / Retention and maturity

Suppose 100 eligible users sign up in a cohort and 35 perform the return event in week four. Week-four retention is 35%. A newer cohort that has not reached week four cannot be treated as having zero returners or compared on that period yet. Subscription customer churn and behavioral inactivity are different constructs.

3 / Candidate value-moment overlap [course example]

In the slide table on PDF p. 12, at least three friends yields 262 users who both met the threshold and were retained, 238 retained users below it, and 140 threshold users who were not retained. Intersection-over-union = 262 / (238 + 262 + 140) = 40.9%. This is the table's largest overlap, about 41%; it is not a causal estimate or a universal three-friend rule.

4 / Multi-factor designs

Two headlines crossed with two images produce 2 x 2 = 4 combinations. That factorial structure can estimate factor and interaction effects with suitable analysis and traffic. The Uber Eats worksheet instead proposes four alternative button experiences with a 25% split each: an A/B/n design. Extra arms increase the demands on sample size and multiple-comparison control.

Connect the branches

Instrumentation > All analyses

Event definitions determine what can be trusted.

Value moment > Active usage

Meaningful activity should reflect recurring value.

Natural frequency > Retention + power curves

Use windows appropriate to the customer job.

Segments > Onboarding + funnels

Different jobs can need different paths and metrics.

Cohort associations > Experiments

A plausible relationship becomes a causal hypothesis.

Retention > Growth + economics

Repeated value supports sustainable growth and lifetime economics.

Qualitative research > Every branch

Investigate the reasons behind observed patterns.

Decision loop: define the user outcome > instrument behavior > locate a problem > explain it with research > test an intervention > evaluate retained value and economics.

Practice: improve repeat ordering

Clarify the decision

Goal: improve repeat customer value, rather than merely buying discounted orders. Clarify the market, target population, observation window and business constraints. Ask whether the problem is discovery, first-order quality, repeat use or pricing.

Measure the journey

Track discovery, menu views, cart, checkout, order placement, cancellations/refunds and delivery. Define activation and recurring activity separately. Validate event coverage, order identity and eligible-user denominators.

Diagnose before proposing

Use entry cohorts and time to a second order; compare platform, acquisition channel and relevant use cases. Inspect funnel leakage and interview users who did not reorder. Treat the workbook's three-day threshold as a hypothesis to validate.

Design the experiment

Hypothesis: a relevant restaurant-specials experience reduces price friction and improves repeat ordering. Randomize eligible users with stable assignment. Define a primary retention metric and guardrails for contribution margin, refunds, delivery quality and unwanted messaging.

Plan inference and decide

Specify the effect worth detecting, traffic, duration and inference plan before launch. Check assignment and instrumentation. Report lift with uncertainty and practical impact. Roll out only if the result supports the stated outcome and guardrails; an inconclusive result needs an uncertainty-based decision.

Use the map for retrieval, not rereading

Rebuild the structure

Hide the atlas and sketch the central topic, five branches and their key subtopics. Compare your reconstruction with the maps. Mark missing links rather than copying everything again.

Explain and apply

For each branch, give a definition, a product example, a measurement choice, an interpretation pitfall and a decision. Answer the embedded interview prompt before opening the answer cue.

Interleave the questions

Mix value, cohort and experiment questions. For example: choose a value proxy, define the cohort and outcome windows, explain confounding, then design a test. This practices choosing an approach rather than recognizing a familiar heading.

Space retrieval

A practical starting schedule is the next day, several days later and about a week later; adapt the spacing to recall and interview timing. This is a suggested routine, not a timing prescription from the repository.

Use the worksheets

Repeat exercises 1-9 on a second product: select it, map value, distinguish activation, identify friction, define activity, specify and analyze cohorts, hypothesize a power-user curve, then write a test plan. Validate assumptions wherever real evidence is available.

Coverage, sources and remaining preparation

Repository snapshot

Reviewed main at commit 6513960a26da26e521527512da5df48043f1874f. The repository contains four PDFs, three RTF resource-link files, a README and a license. The RTF files point to slide/workbook copies and a student community; the study content summarized here comes from the included PDFs. External linked copies and video lessons were not treated as additional source content.

Course slides - 42 pages. Foundations pp. 3-5; value pp. 6-17; performance pp. 18-28; segmentation pp. 29-35; UX/testing pp. 36-42.

Blank worksheet - 10 pages. Nine exercises; used as the practice sequence.

Uber Eats worksheet - 10 pages. Completed illustrative exercises; hypotheses and proposed criteria are not outcomes.

IDEO brainstorming - One page. Seven ideation principles.

Additional interview preparation [outside repository scope]

The materials are introductory. Depending on the role, add SQL and event modeling; sample-size and statistical-test practice; causal inference; experiment interference and sequential methods; and SaaS economics such as MRR/ARR, NRR/GRR, CAC and contribution-based LTV. The [extension] sections improve precision but do not constitute a full treatment of these subjects.